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RESEARCH ARTICLE

Artificial Flora Algorithm-Based Feature Selection With Support Vector Machine for Cardiovascular Disease Classification

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ABSTRACT Accurately categorizing medical information is crucial for determining effective cardiac treatment options, especially as the volume of data grows and feature selection becomes increasingly challenging. This work proposes a model to identify the presence of Cardiovascular Disease based on various patient features, aiming to enhance prediction accuracy through a powerful feature selection method. This approach utilizes the Cleveland dataset by combining the Artificial Flora Optimization algorithm with the Support Vector Machine. The proposed algorithm functions as a meticulous gardener, selectively identifying the most significant features for heart disease prediction through an objective function. The model demonstrates impressive performance, achieving an accuracy of 96.63%, specificity of 95.73%, sensitivity of 97.74%, precision of 94.89%, and an F1-score of 96.29%. The model promises high-accuracy heart disease predictions by optimizing feature selection, potentially transforming clinical practice, and advancing research. The novel combination of the proposed technique holds significant potential for improving medical categorization and patient outcomes.

INDEX TERMS Artificial flora optimization, fitness function, feature selection, ROC analysis, support vector machine, cardiovascular disease.

I. INTRODUCTION

CardioVascular Diseases (CVDs) kill 17.9 million people each year, making them the leading cause of mortality worldwide. CVDs are heart and blood vessel problems that include conditions including "coronary heart disease", "cerebrovascular disease", "rheumatic heart disease", and others. More than four out of every five CVD deaths are caused by heart attacks or strokes, and one-third of these deaths occur before the age of 70.

The most significant behavioral risk factors for heart disease and stroke are poor diet, inactivity, tobacco use, and alcohol abuse. People may develop high blood pressure, high blood glucose, high blood lipids, overweight, and obesity as a result of behavioral risk factors. These "intermediate risk factors" can be evaluated in primary care settings and indicate an increased chance of complications such as heart attack, stroke, or heart failure. Quitting smoking, lowering salt intake, eating more fruits and vegetables, exercising frequently, and avoiding excessive alcohol consumption have been shown to reduce the risk of cardiovascular disease. Health policies that support environments where healthy options are affordable and accessible are vital if people want to learn and maintain healthy habits.

Premature deaths can be avoided by ensuring that the patients receive the appropriate care. Additionally, it is thought that exhaustion and chest discomfort are the two signs of heart disease that people are most familiar with [1]. Modern computational researchers have developed several feature selection approaches to address this problem [2], [3], [4]. This paper covers the prediction of heart disease and recent breakthroughs in feature selection.

The feature subsets are ranked using a simple heuristic evaluation function in correlation-based feature selection [5], [6]. This heuristic assessment tool efficiently discovers the more significant qualities due to their high correlation

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values. Therefore, features with low correlation are removed from the training dataset as well as the testing dataset [7]. "Correlation-based feature selection" strategies are also utilized to delete duplicate features from the datasets. Using a "consistency filter-based feature selection technique", a subset S is also selected from the complete collection of features [8]. The best position is updated based on a comparison between S's feature count and the present best feature count. The chosen features are then rendered using a variety of feature selection techniques, including Information Gain, ReliefF, and others [9], [10].

Information gain is one of several feature assessment procedures that are commonly utilized in a wide range of applications. This assigns a numerical value to each feature depending upon a user-specified threshold. Entropy is a concept that is incorporated into Information Gain for ranking the attributes. The expected value for the Information Gain technique is computed using the combined evidence of the objective variable [11], [12], [13]. Artificial intelligence and machine learning approaches have been developed recently for the diagnosis of various diseases. For instance, early detection of heart illness using machine learning techniques is discussed. In the healthcare sector, the Case-based reasoning technique is suggested for medical diagnosis, outlook, and suggestion. Similar to this, Manogaran et al. [14] forecasts coronary artery disease using a Genetic algorithm (GA) and binary particle swarm optimization. The healthcare dataset is also used by [15] to extract hidden values and insightful data using a range of classification and regression techniques.

These days, there is so much data being produced that it exceeds all of the data produced throughout the history of the human species [16]. Therefore, it is imperative to find the best techniques for managing these vast amounts of data. Intelligent bio-inspired algorithms can tackle dynamic issues with limited and inadequate knowledge [17]. These algorithms are said to as intelligent since they can adapt and assimilate just like living things. However, a significant number of bio-inspired algorithms have been overlooked due to the field's quick development. Most of the widely used are Genetic algorithms and Particle Swarm Optimization (PSO) [18], [19]. However, the existing studies have some inconsistencies.

- Most publications focus on the correlation between features and labels, as well as feature redundancy, but not on feature performance. This ignores the contribution of features to classification, resulting in the loss of feature information.
- The process for selecting significant and informative features, as well as removing redundant features, is unclear.
- The existing efforts are insufficient to address real-life situations. A large amount of theoretical as well as practical work has been done in a few bio-inspired algorithms like Genetic Algorithm and Particle Swarm Optimization. However, bio-inspired algorithms like Artificial Flora Optimizarion (AFO) have limited literature available.

This work introduces a novel feature selection method based on the Artificial Flora Optimization with Support Vector Machine (AFO-SVM) algorithm. The study covers both theoretical and practical elements of AFO. AFO was used to extract features and then coupled with other machine learning algorithms like SVM to optimize features for detecting heart illness. AFO can forecast indicators such as sugar and cholesterol levels. The typical AFO algorithm cannot determine the appropriate method for updating the plant's location and propagation distance. To overcome this, we developed a special function that determines the optimum position. As a fitness function, the suggested AFO-SVM technique employs a constant value SVM classifier to reduce the number of features while increasing accuracy. The main contributions of the work are as follows.

- Investigating the effectiveness of feature selection using AFO and AFO-SVM methods.
- Empirical results show that the proposed AFO-SVM selects the most optimal and significant characteristics compared to existing methods.

The remaining work is organized as follows. The related work is presented in Section II. This section reviews studies that use machine learning and bio-inspired computing to detect heart disease. Section III describes the proposed methodology. Section IV includes the performance evaluation. Section V describes the results and discussion, Section VI highlights the conclusion and recommendations for future work.

II. RELATED WORK

Ordonez [20] makes an excellent concept for a rule-reduction algorithm. This method uses real-time medical records to detect heart attacks in their early stages. The number of "association rules" used in the "mining process" are reduced with the help of this strategy. As a result, output criteria of the algorithm are used to precisely classify cardiac disease. The performance evaluation criteria are also used to evaluate the effectiveness of the generated rules [21]. The generated parameters are computed for a predetermined set of examples. These rule-generation algorithms have been used to predict cardiac events early on with success in recent years.

Tan et al. [22], used wrapper method-based genetic algorithms to provide a novel feature selection methodology. By finding the more significant traits that matter, this technique seeks to effectively categorize heart disease independently. The fitness function is used to sort a set of provided features to decide which features are most appropriate. Similarly, Nahar et al. [23] created a groundbreaking method for feature extraction based on a number of "meta-heuristic algorithms" such as Tertius and Predictive Apriori. It is possible to classify disease using the generated features with an accuracy of 87%. Using performance evaluation parameters, it is demonstrated that the recommended metaheuristic-based feature selection algorithm performs well [24].

Polat [25] breakthrough "Kemel F-score Feature Selection (KFFS)" method, developed in 2009, is used to eliminate unnecessary and redundant characteristics and choose the most important features. Proposed method is tested on huge electronic health record, and the algorithm's effectiveness is measured using parameters such as "sensitivity", "accuracy", "precision", "specificity", "true positive rate", and "false positive rate" [26]. The "Radial Basis Function (RBF)" and F-score approaches are used to create the two types of kernel functions that make up the proposed DFFS methodology. The RNF kernel approach's main role is the conversion of characteristics from medical datasets into a kernel space [27], [28]. The first version of the suggested framework used the F-score technique to determine the high-dimensional F-score values.

Furthermore, Luukka [29] adopts a differential evolution classification based on "Principal Component Analysis" (PCA) to provide more effective rules for the early prediction of heart disease. The construction of the model and the removal of duplicate nonsensical features from both the training and testing datasets are the two key goals of this PCA approach. They assessed the effectiveness of the proposed technique using the standard Electronic Medical Record (EMR). The patient's symptoms and behaviour, heart rate, blood pressure, myocardial infarction are only a few of the patient demographical aspects that are considered when predicting diabetes Muthukaruppan [30].

Yan et al. [31] suggested a unique feature selection strategy based on genetic algorithms to find the most relevant risk factors for heart disease. Several heart disease datasets were used to test this technique. In addition to the techniques already mentioned, a variety of AI procedures and processes are employed to efficiently forecast and categorize heart diseases Koloseni [32]. These techniques are also used to determine the most significant traits. Machine learning, artificial intelligence, genetic algorithms with "binary particle swarm optimization (BPSO)", and "case-based reasoning" techniques were all employed to forecast heart illnesses.

By raising the neural network's heart disease, this technique enhances its performance by roughly 10%. The experimental findings are compared to the starting weights. The evolutionary algorithm performs this weight-changing mechanism, which provides improved weights for the neural network. On the Z-Alizadeh Sani dataset, Tsipouras et al. [33] proposed a method with accuracy, specificity, and sensitivity values of 93.85, 97, and 92%, respectively. For heart disease diagnosis, [28] introduced the "kernel extreme learning machine (KELM)" and "improved grey wolf optimization (IGWO)" approaches. The IGWO attempts to extract the best features from the dataset. To evaluate the risk variables for heart disease, Vivekanandan and Iyengar [38] suggested a modified differential evolution (DE) technique. The method was evaluated using a dataset comprising a variety of heart illnesses, including "hypertension", "chronic pulmonary illness", "congenital heart disease", "rheumatic valvular heart disease" and "coronary heart disease".

Jabbar et al. [39] developed a classification algorithm that combines "chi-square" and "random forest" to predict cardiac disease. The effectiveness of the suggested approach is examined using heart disease data sets from UCI. When compared to previous methodologies, the experimental findings show that the proposed strategy provided superior classification accuracy. Paul et al. [40] proposed a new 'fuzzy decision support system' to predict cardiac disease based on the 'genetic algorithm'.

Inbarani et al. [41] proposed a novel supervised feature selection approach with Particle Swarm Optimization hybridization. The performance of this method is evaluated using a variety of performance indicators such as 'ROC analysis', 'specificity', 'true positive rate', 'sensitivity' and 'false positive rate'. To validate the proposed technique, a huge electronic health record is deployed. For the early diagnosis of heart problems, a feature selection based on Least Square Twin Support Vector Machine (LSTSVM) is proposed by Tomar and Agarwal [42]. The F score statistics of this method are utilized to calculate the weight of each feature. Risk factors for cardiovascular disease are detected using this suggested LSTSVM.

Elhoseny et al. [43] presented a cutting-edge automated heart disease diagnostic system (AHDD) that bridges a binary convolutional neural network (CNN) with a unique multiagent feature wrapper model (MAFW). The MAFW leverages four software agents powered by GA, SVM, and Naïve Bayes (NB) algorithms. These agents collaborate to guide GA in a comprehensive search for relevant heart disease characteristics and refine the weights of the SVM and Naive Bayes models for initial classification. The system then fine-tunes CNN to integrate the optimal feature set, ultimately aiming for highly accurate heart disease identification.

Enad [44] made a comprehensive review that explores the potential of AI to revolutionize the diagnosis of heart disease. It critically evaluates existing AI-based methods for their efficiency and limitations, not only in diagnosis, but also in broader prevention efforts. By analyzing state-of-the-art advancements in AI, including machine learning, Deep Learning (DL), and even quantum computing, this review lays the groundwork for future research and development. Its ultimate objective is to enable the application of AI solutions for earlier detection and improved patient outcomes in cardiac diseases.

Sarra et al. [45] have conducted a study that tackles the challenge of limited and unbalanced data in Heart Disease (HD) diagnosis by proposing two novel DL frameworks: GAN-1D-CNN and GAN-Bi-LSTM. These frameworks leverage Generative Adversarial Networks (GANs) to create synthetic samples, augmenting datasets, and boosting the performance of subsequent 1D-CNN and Bi-LSTM models for accurate HD prediction. The study rigorously evaluates both frameworks in three unbalanced datasets in the real world, using metrics such as accuracy, specificity, and AUC, demonstrating their potential to improve heart disease diagnosis in clinical practice.

The study by Paudel et al. [46] found that Light Gradient Boosting Machine consistently outperformed other algorithms in terms of training accuracy. While all algorithms exhibited similar precision, recall, and F1-score levels, AdaBoost lagged. Furthermore, an analysis of feature importance revealed that ''kcm'' and ''troponin'' were consistently identified as critical factors in predicting heart attacks across all models. Table 1 presents the comparison with the state-of-the-art methods.

Ali and Mohammed [47] proposed a study that reviews and assesses the cutting edge of omics-related AI research, showcasing the immense potential of AI for omics data analysis while critically identifying key challenges that must be tackled to unlock its full potential. By systematically gathering and analyzing research on the applications and challenges of omics-AI, this review goes beyond existing studies, offering unique insights into this rapidly evolving field. By addressing crucial issues such as preprocessing, data sets, model validation, and testbed applications, this review paves the way for future research to fully unlock the transformative potential of AI in omics data analysis and its subsequent impact on clinical applications.

AFO-SVM uniquely combines the efficiency of AFO with the classification strength of SVM to address challenges such as computational inefficiency, premature convergence, and hyperparameter sensitivity that are common in other state-of-the-art algorithms. This biologically inspired optimization framework provides a fresh perspective, particularly in scenarios involving high-dimensional or noisy data.

It often involves fewer parameters and operations than traditional evolutionary algorithms like Genetic Algorithms. GAs and PSO often require complex operations (e.g., crossover, mutation, or velocity updates) that increase computational cost. AFO's simplicity can lead to faster convergence and reduced overhead.

III. METHODOLOGY

The proposed methodology is used to accurately classify whether the person has cardiovascular disease or not. Different feature selection techniques and ML-based algorithms are tested for their ability to accurately diagnose CVD using both entire and selective features. A qualitative system for predicting CVD is then proposed. The suggested strategy uses multiple ML classifiers to improve prediction accuracy. Our suggested approach employs four steps. (1) Preprocessing of heart disease dataset, (2) A feature selection step involving the Artificial Flora Optimization with Support Vector Machine (AFO-SVM). Support vector machine based objective function is constructed in this study. This function is utilised in the AFO to pinpoint important heart disease risk factors. (3) Classification using support vector machine. The classification is performed on the dataset which is an output from the second step, (4) Evaluation of classification algorithms using many evaluation metrics. Fig. 1 depicts this paper's overall workflow.

A. ARTIFICIAL FLORA-BASED FEATURE SELECTION

The goal of bio-inspired algorithms is to quickly focus on the best option. When compared to typical deterministic algorithms, the bio-inspired algorithm outperforms. Ideal solutions are rapidly classified using bio-inspired algorithms. The offspring are chosen using a random manner in this method. Additionally, the offspring in AFO travel at random from source to destination. To find the optimum solution, a fitness function is computed for each movement and iteration. One of the most crucial AFO methods is determined to be the fitness function. Based on how well the fitness function works, the AFO algorithm's efficacy and efficiency are calculated. Based on their fitness value, the best offspring are chosen by this algorithm.

1) BIOLOGICAL FUNDAMENTALS

Plants have many ways of dispersing their seeds. There are two types of seed dispersal: autochory and allochory. Autochory denotes that plants spread through external influences, whereas allochory refers to plants that spread with their assistance. Plants can migrate autonomously to the ideal, suitable habitat thanks to autochory. Allochory, on the opposite hand, fosters plant migration to remote and unknown areas. For example, based on variations in wind speed and direction, the wind within an anemochory can determine the seed distance and spread direction. These techniques of propagation decrease the likelihood of plant extinction while broadening the scope of flora inquiry.

The distribution range of the flora is declining, shifting, or expanding as a result of changes in the climate, natural habitat, or strong competition. One type of plant can perfectly evolve when it is transported to a new location. As a result, flora migration alters its geographic range while also causing the flora to evolve, go extinct, and then reappear. A plant cannot move without brains, but through seed dispersal and sexual reproduction, plants can find better places to live. The parent plant disperses seeds randomly across a predetermined distance in order to facilitate flora migration and reproduction. A seed's chance of survival can change depending on its environment. When a plant is ready and in an environment that is suitable, it endures and disperses seeds. Allochory creates the possibility that, before a single region's flora completely vanishes, it will spread out across the remaining areas. Multigenerational propagation is how flora spreads to the right location. Flora is able to fulfill its purpose of identifying the optimal growth circumstances in a habitat through its processes of migration and reproduction cause evolution, extinction, and rebirth.

Because of extreme competition, shifting climatic conditions, or changes in the natural environment, the flora is expanding, moving, or decreasing in its distribution area. One type of plant can perfectly evolve when it is transported to a new location. As a result, flora migration alters its geographic range while also causing the flora to evolve, go extinct, and then reappear. A plant cannot move without

TABLE 1. State-of-the-art methods: contributions and research gaps.

Ref	Dataset	Optimization Algorithm	Number of features selected	Accuracy (%)	Research gap
[15]	Cleveland Statlog	Genetic Algorithm	13	87 86	The proposed model cannot handle complex feature interactions and non-linear relationships in real-world data.
[18]	Cleveland	GA-SVM	7	88	Testing the model on diverse datasets leads to the lack of robustness and applicability of the model
[19]	Cleveland	Improved PSO	6	84.36	Optimal weight identification in improved PSO increases time complexity.
[23]	UCI	Metaheuristic	13	87	Limited Feature selection Inconsistent Performance Computationally expensive.
[33]	Real-time dataset from the University Hospital of Ioannina	Optimized fuzzy model	19	85	Need to address class imbalance as demographic details are also included.
[35]	Heart Disease	Black Hole Optimization (BHO)	7	92.3	Optimal weight identification in improved PSO increases time complexity.
[36]	Cleveland	QPSO	13	92.81	Integrating the model into clinical practice requires careful considerations such as data quality and computational cost.
[37]	Cleveland	Artificial Bee Colony (ABC)	8	86.76	The accuracy of the model heavily relies on the quality of the selected features
[40]	IEEE Dataports	Genetic Algorithm	12	79.5	Random sampling is used to reduce the number of instances which can lead to bias and certain subgroups of data may be under-represented or overrepresented.

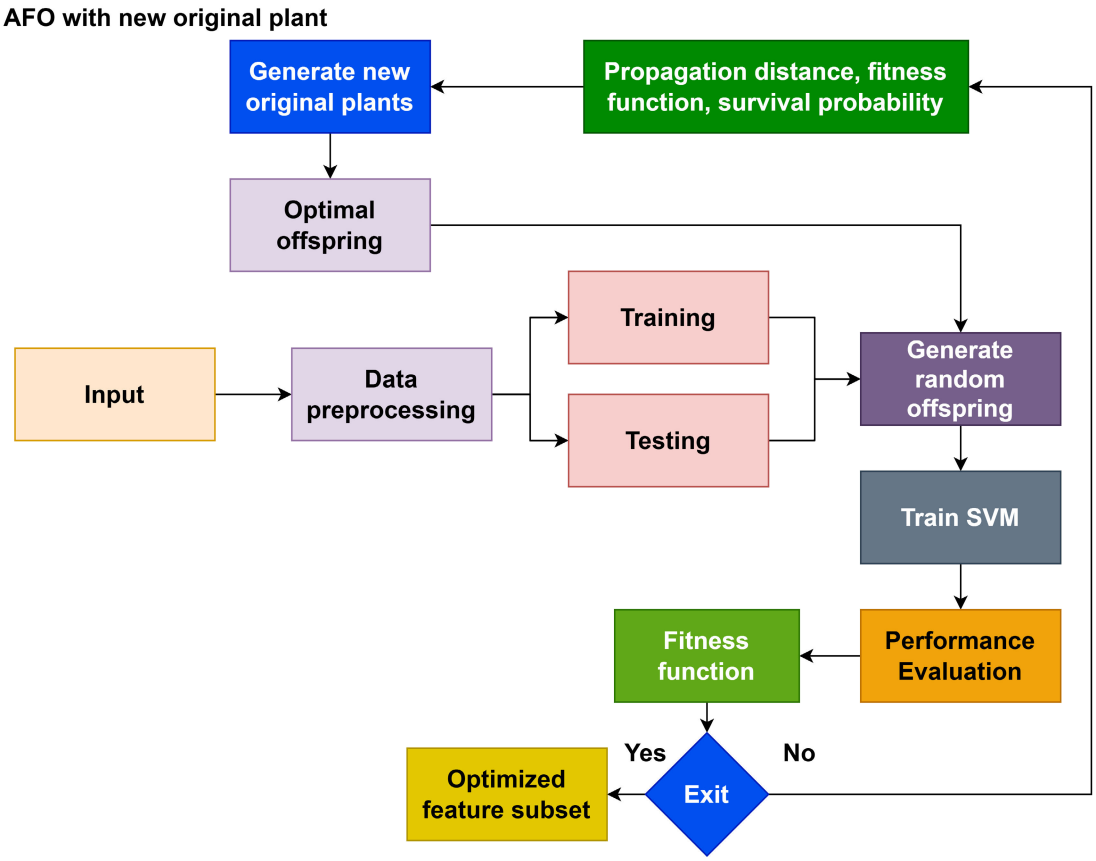


FIGURE 1. Workflow of the proposed methodology for CVD prediction.

brains, but through seed dispersal and sexual reproduction, plants can find better places to live. The parent plant disperses seeds randomly across a predetermined distance in order to facilitate flora migration and reproduction. A seed's chance of survival can change depending on its environment. When a plant is ready and in an environment that is

suitable, it endures and disperses seeds. Allochory creates the possibility that, before a single region's flora completely vanishes, it will spread out across the remaining areas. Flora uses multigenerational propagation to travel to the appropriate place. Flora's migration and reproduction cause evolution, extinction, and rebirth allow it to complete its mission of determining the environment's best conditions for growth.

2) ARTIFICIAL FLORA ALGORITHM THEORY

Four fundamental categories—the original plant, its offspring, its location, and its propagation distance have been established. Original plants are those that are actively dispersing seeds. Original plant seeds from offspring plants cannot quickly disperse seeds. "Location" is a reference to the plant's location. The distance that seeds will travel is referred to as the "propagation distance". Selection behaviour, spreading behaviour, and evolution behaviour are three important behavioural patterns.It is possible for a plant to evolve to match the characteristics of its surroundings. Allochory or autochory-based seed movement is referred to as spreading behavior. Select behavior describes how the environment affects flora's ability to survive or perish. The social behaviors listed above are made simpler by the following idealized circumstances: The rules for selecting the offsprings are listed in Table 2.

Fig. 2 illustrates the migration and reproduction process of the AFO algorithm. Red lines indicate migration paths, blue lines represent local exploration or pollination, and green lines signify reproductive interactions. Colored circles denote the search spaces or zones of influence for different flora agents.

- According to Rule 1, there were no specific species in the area. Original plants had dispersed over arbitrary locations in a given region as a result of rapid changes in the environment or specific artificial action, as seen in $\Diamond(X1)$in Fig. 2.
- According to Rule 3, original plants disperse seeds within the propagation range. The offspring is positioned in the blue circle in Fig. 2, Distance1, representing the propagation distance of $\Diamond(X1)$ units. Offspring showed as $\triangle(a1, a2, a3)$ in Fig. 2.
- The fitness is represented with the number of $\triangle$. According to Fig. 2, a higher presence of $\triangle$ determines a fitter offspring. Fitness will be higher if progeny remains close to the mother plant: $Fitness(a1) > fitness(a2) > fitness(a3)$, which is equivalent to Rule 5.
- According to Rule 4, some offspring survive because of fitness differences. As seen in Fig. 2, a plant that is alive is represented by a solid line, and a dotted line represents a plant that is not alive. The fittest offspring, a1, did not survive because of competition and other factors. However, a2 with a lower fitness level than a1 will live and regenerate into a new original plant.
- In Fig. 2, a new original plant can disperse seeds, much like the illustration of $\square(b1, b2, b3)$, where b1 and b3 can both survive while b2 cannot. In order to create the newest original plant, one plant will be picked at random from b2 and b1, whereas b1 can be chosen as shown in Fig. 2.
- Distances 2 and 3 are known as $\triangle(a2)$ and $\square(b1)$ propagation distances, respectively. According to Rule 2, $Distance_2$ develops from $Distance_1$, however, $Distance_3$ can be learned from both $Distance_2$ and $Distance_1$. When b1 disperses seeds, the distance of b1's descendants will rely on both $Distance_3$ and $Distance_2$.
- Plants frequently release seeds into the environment, causing flora to migrate after it has found the greatest place to survive.
- Allochory will randomly produce a new original plant within the region if the entire offspring plants, as shown in Fig. 2 by $\nabla(C1, C2, C3)$, are unable to be alive.

TABLE 2. Conditions of AFO algorithm.

Rules	Details
1	Some species may become randomly distributed across a region devoid of such species or become the most basic original plant as a result of an abrupt environmental change or specific artificial activity.
2	As the environment changes, plants evolve to adapt to the new conditions. As a result, the propagation distance of offspring plants cannot be viewed as the full inheritance of a parent plant but rather evolves based on the distance between plants of a previous generation. In addition, in a perfect world, children only learn things from their nearest two generations.
3	In this ideal scenario, the range is a circle with a maximum propagation distance radius while the initial plant autonomously disperses seeds. Plants reproduce and disperse over the circumference of a circle.
4	Climate and other environmental elements, such as temperature, might vary depending on where you are. Plants thus have different survival probabilities. Survival likelihood can be connected with plant fitness in the given environment. In this context, plant environment adaption is referred to as fitness. Fitness, thus, can be defined as the likelihood of a plant surviving in its environment. The higher your fitness level, the higher your chance of surviving. The inter-specific competition also results in the death of plants with high fitness.
5	The likelihood of survival can decrease with increasing distance from the source plants. Where the offspring plant stays apart from the original plant of the same generation, it may be greater due to differences between the prior environment and the current environment.
6	Although seeds can travel externally, their propagation distance will never exceed the area with the maximum limit due to restrictions, such as animal activity range.

3) REPRESENTATION OF AGENTS

Each feature subset in Artificial Flora-based feature selection could be seen as the location of a certain flora. There were N features (flora) in each individual subset, where N was the total number of features/solutions in the original set.

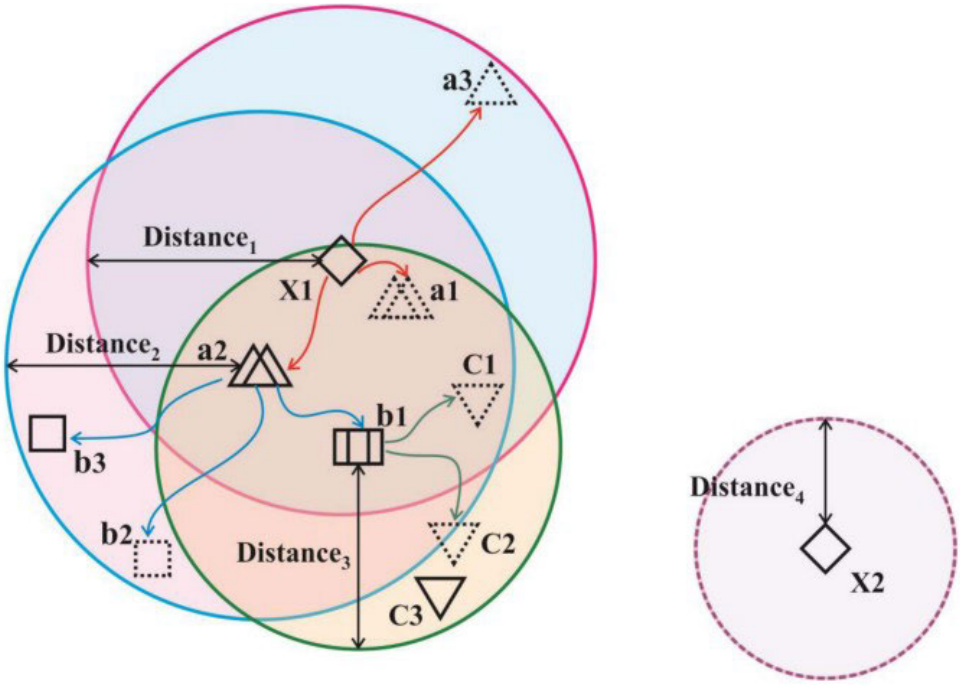


FIGURE 2. Processes of fundamental migration and reproduction in the artificial flora algorithm.

4) POPULATION INITIALIZATION

In a previous stage, AFO created an original flora at random using N solutions, where N plants were present.

B. EVOLUTION BEHAVIOR

Original plant seeds are dispersed within the radius of the propagation distance. The propagation distance can be derived from the propagation distances of the parent and grandparent plants.

$$d_j = d_{1j} \times \text{rand}(0, 1) \times C_1 + d_{2j} \times \text{rand}(0, 1) \times C_2 \quad (1)$$

Here, d_{1j} signifies the distance between the parent and grandparent plants, and d_{2j} the distance between the grandparent and parent plants. In contrast to an independent uniformly distributed number within the range of (0,1), the learning coefficient is denoted by the letters c_1 and c_2, respectively.

Distance of new grandparents' propagation

$$d'_{1j} = d_{2j} \quad (2)$$

The standard deviation between the positions of the original plant and the offspring plant is the new grandparent's propagation distance.

$$d'_{2j} = \sqrt{\frac{\sum_{i=1}^{N} \left(P_{i,j} - P'_{i,j}\right)^2}{N}} \quad (3)$$

C. SPREADING BEHAVIOR

The artificial flora method can initially generate original flora at random using N solutions, each of which has N plants. $P_{i,j}$ is a matrix that can be used to express the original plant position. Here, the number of plants within the flora is denoted by j, whereas the dimension is denoted by i.

$$P_{i,j} = \text{rand}(0, 1) \times d \times 2 - d \quad (4)$$

In this case, the highest limit is denoted by the letter d, and the array of random numbers with a uniform distribution between (0,1) is denoted by the notation rand(0,1).

Depending on the role of propagation, the position of the offspring plant can be produced:

$$P'_{i,j\times m} = D_{i,j\times m} + P_{i,j} \quad (5)$$

Here, "m" stands for the number of seeds that one plant produces, "$P_i, j \times m$" stands for the location of the offspring plant, "P_i, j" stands for the original plant, and "Di, $j \times m$" stands for a random number having a Gaussian distribution, with mean 0 and variance d_j. Equation 4 will be used to develop a new original plant whenever an offspring plant cannot survive.

D. SELECT BEHAVIOR

The condition of the plants' offspring can be assessed using the survival probability:

$$p = \left| \sqrt{\frac{F\left(P'_{i,j\times m}\right)}{F_{\max}}} \right| \times Q_x^{(j\times m-1)} \quad (6)$$

Here Q_x refers to selective probability, Q_x to the power of (j × m − 1) meant by $Q_x^{(j\times m-1)}$, its value, which is interpreted as the fitness of the offspring plant, is constant between

0 and 1. It keeps its distance from the original plant low. The algorithm's capacity for exploration is measured by Q_x. In order to obtain a local optimal solution, Q_x must continue to be the larger of the two problems. Maximum fitness within a generation of flora is given by F_max, whereas fitness of the j-th solution is denoted by $F\left(P'_{i,j\times m}\right)$.

E. ANALYSIS OF THE PROPOSED ALGORITHM'S FLOW

The objective function is mentioned in the equation of fitness. It is possible to predict whether or not the offspring plant will survive based on the result of the roulette wheel selection. With "accept according to probability" as its primary objective, the roulette wheel selection process can be referred to as a proportion choice strategy; there are many choices, each of which carries a potential score. Additionally, the score value will never solely determine selection. Accepting probability as a basis for selection is possible. A higher score will increase the likelihood of acceptance.

To ensure the survival of the plant's descendants, a random number 'r' with a uniform distribution of [0, 1] on each try is produced. To survive, survival probability P needs to be higher than r. Select N offspring plants to act as new actual plants that reproduce the above mentioned behaviours with the necessary accuracy from among the survivors. It is probable or the highest number of repetitions possible. Fig. 3 displays flowchart for modified Artificial Flora Optimization method. Crucial phases for the AFO algorithm:

- Create N original plants and use Eq. (4) to initialize the population.
- Calculate the propagation distance using Eq. (1), (2), and (3).
- Using Eq. (5), produce progeny plants and assess their fitness.
- Use Eq. (6) to determine the probability that plants' progeny will survive; use the roulette wheel selection procedure to determine whether offspring will survive or not.

F. ROULETTE-WHEEL SELECTION ALGORITHM

In this selection, a probable estimate of an offspring's fitness is divided by the original fitness of the population. The notion that shape and fitness are inversely connected informs the size of the roulette- wheel. Each child, thus given a piece of the roulette wheel. The population's total number of plants determines how the roulette wheel rotates. Every time, a seedling is selected for the next generation from a pool of prospective plants. The following Algorithm 1 basically illustrates how a roulette wheel is selected.

Algorithm 1 Roulette-Wheel Selection Algorithm

Step 1: Compute the total expected value of the population, denoted as t.

Step 2: Repeat N times:

1. Pick a random integer r such that r lies between 0 and t.
2. For each individual i in the population:
   a. Compute the cumulative sum of the expected values until the sum exceeds or equals r.
3. The individual whose cumulative sum exceeds or equals r is selected.

G. SUPPORT VECTOR MACHINE (SVM)

Vapnik suggests using SVM for classification, regression, and density estimation problems. SVM's goal is to identify the data points $x_i$that correspond to a certain decision rule $g(x) = sign(wx + b)$ on the basis of the hyperplane $wx + b = 0$ [48]. This hyperplane is recognized by the SVM as being the farthest from the data points x_i, or $+b = 0$. When using new data, the core concept of the SVM is that the distance of a hyperplane from any data points would reduce the possibility of erroneous classification outcomes.

Let's say a pattern contains a pair

$$\{(x_i, y_i)\}_{i=1}^{N} \quad \text{where} \quad x_i \in \mathbb{R}^n,\ y_i \in \{-1, 1\}$$

Let us assume that the space of labels is $y \in \{-1, 1\}$ and the space of patterns is $x \in \mathbb{R}^n$ SVM's goal is to identify a classifier $y(x)$

$$y(x) = \text{sign}\left[\sum_{i=1}^{N} \alpha_i y_i K(x_i, x) + b\right] \tag{7}$$

where b is real constant, and α_i is positive real constant. Frequently, the Kernel function

$$K(x_i, x) = \phi(x_i) \cdot \phi(x) \tag{8}$$

where $\phi(x)$ denotes a non-linear high dimensional mapping, <.,.> denotes inner product operation. Let's assume that a linear hyperplane divides the input data set. In high-dimensional space, the separation of this process is found by the following equation:

$$y_i\left[\mathbf{w}^T\phi(x_i) + b\right] \geq 1, \quad \text{where} \quad i = 1, \ldots, N \tag{9}$$

When a splitting hyperplane does not exist, a slack variable n is introduced

$$y_i\left[\mathbf{w}^T\phi(x_i) + b\right] \geq 1 - \xi_i,$$

where

$$i = 1, \ldots, N\xi_i \geq 0, \quad i = 1, \ldots, N \tag{10}$$

According to structural risk minimization theory, minimization is defined as

$$\min_{\mathbf{w},\xi} J_1(\mathbf{w}, \xi) = \frac{1}{2}\mathbf{w}^T\mathbf{w} + c\sum_{i=1}^{N}\xi_i \tag{11}$$

On the foundation of the Lagrangian function, the optimal point is calculated as

$$\max_{\mathbf{w},\beta}\ \min_{\mathbf{w},b,\xi} L_1(\mathbf{w}, b, \xi, \alpha, \beta) \tag{12}$$

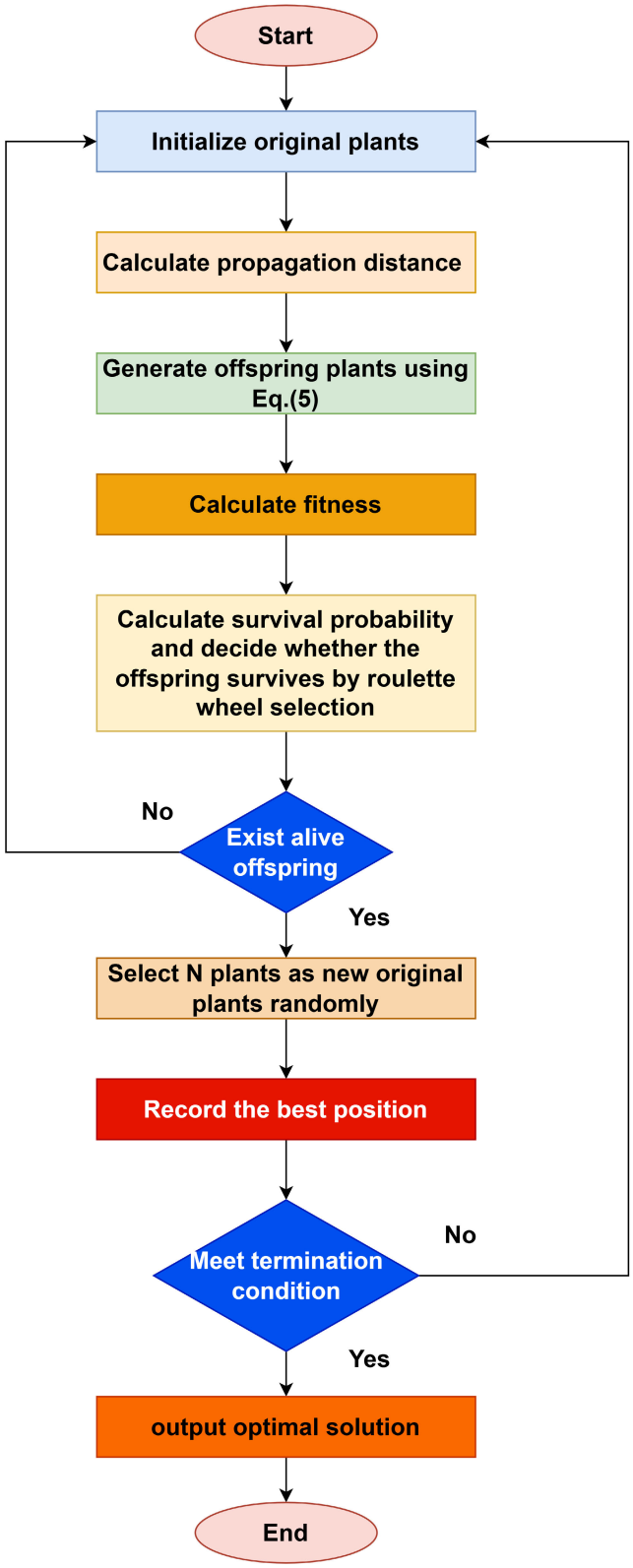


FIGURE 3. Algorithm flow of modified artificial flora optimization algorithm.

The partial differentiation with zeros is applied in

$$\frac{\partial L_1}{\partial b} = 0, \quad \mathbf{w} = \sum_{i=1}^{n} \alpha_i y_i = 0 \tag{13}$$

By using Eq. (12) in (6), quadratic programming (QP) will result.

$$\min_{\alpha} Q_1(\alpha) = \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j y_i y_j K(x_i, x_j) \tag{14}$$

By interpreting the aforementioned QP problem Eq. (14), taking into account the limitations stated in Eq. (13), As shown in Eq. (8), a hyperplane is found on a high-dimensional space and a classifier is found in original.

The AFO algorithm optimizes feature selection for heart disease prediction by identifying the best subset of features that improve classifier accuracy. This iterative process starts with a randomly generated population of candidate feature sets and progressively refines them through communication and survival probability selection. Notably, AFO evaluates each candidate based on both maximizing accuracy and minimizing feature count, leading to a more efficient model. When tested on the Cleveland dataset, AFO successfully selected key features that, when fed into an SVM classifier, achieved a remarkable 91.46% accuracy, surpassing the 79.35% achieved using all features. This demonstrates the effectiveness of AFO in identifying crucial features for accurate heart disease prediction while reducing model complexity.

The Algorithm for the proposed AFO algorithm with SVM classifier for feature selection is shown in Algorithm 2.

The feature selection is based on the optimization function. The optimization function $f(I)$ depends on two other functions namely $f_1(I) and f_2(I)$. The first function is the classification accuracy of SVM computed for each subgroup of the population created in each iteration. The second function is the number of selected features for each iteration. Then these two values are used to compute the optimization function with the help of a parameter α. The value of α depends on the propagation distance which is an important parameter in AFO algorithm. The Initial value of propagation distance is taken as 0.5 times of initial population size. Generally, a larger propagation distance will yield a better solution. α is greater than 0.5 and less than 1 for longer distance and less than 0.5 for shorter distance off springs. After calculating the fitness function, the procedure is repeated for 10N times, N is the number of features. The feature having the minimum fitness value is selected as a important feature among the other features.

Table 3 outlines key parameters and their values for a heart disease prediction model using the AFO-SVM algorithm. The model starts with 13 original features and employs AFO-SVM to select the most relevant 6 features. Feature scaling is performed to ensure features have similar ranges, and the regularization parameter is set between 0 and 1 to balance model complexity and generalization. A linear kernel is chosen for the SVM, indicating a focus on capturing linear relationships in the data. The model runs for 10 iterations

Algorithm 2 Artificial Flora Optimization Algorithm Using SVM for Feature Selection

Step 1: Input the training and testing datasets.

Step 2: Pre-process the input data using Z-Score Normalization:

$$Z = \frac{x - \mu}{\sigma}$$

where σ is the standard deviation and μ is the mean.

Step 3: Create a random population of n offspring.

Step 3.1: Determine the optimization function $f(x)$ for every x in the population.

Step 3.2: Train the SVM classifier using the training dataset.

Step 3.3: Construct the optimization function, including:

Fitness Function:

$$f_1(I) = \frac{TP + TN}{TP + TN + FP + FN}$$

where:

TP=True Positive; FP=False Positive; TN=True Negative; FN=False Negative

Number of Selected Features:

$$f_2(I) = \left(1 - \frac{m_r}{p}\right)$$

where:

- m_r is the number of offspring having a value of 1
- p is the length of the offspring

Optimization Function:

$$f(I) = \alpha f_1(I) + (1 - \alpha) f_2(I) \quad \text{subject to} \quad 0 < \alpha < 1$$

Allow $\alpha > 0.5$ for greater distance offspring and $\alpha < 0.5$ for shorter distance offspring.

Step 4: Perform operations to calculate propagation distance and generate random plants.

Step 5: Terminate the procedure if one of the following conditions is met:

- The process reaches $10N$ iterations, where N is the number of features.

Otherwise, return to Step 2.

TABLE 3. Parameters of proposed AFO-SVM algorithm.

Parameter	Value
Model	AFO-SVM
Input features	13(original features)
Output features	6(selected features by AFO-SVM)
α parameter (Feature scaling)	Lies between 0 and 1
Number of flora (NF)	Random search
Propagation distance	Random search
Number of iterations	10N (10*Number of original features)
Kernel type	Linear
Regularization parameter	Lies between 0 and 1

per original feature, providing ample optimization time. Interestingly, the number of flora (NF) and propagation distance (PD) in AFO aren't fixed but determined through random search, suggesting a strategy to explore different parameter combinations for optimal performance.

The comparison between AFO-SVM and other methods depends on the specific data, task, and desired model characteristics. AFO excels in balancing accuracy and feature count while adapting to feature importance, making it potentially suitable for scenarios where model complexity and efficiency are concerns. However, its complexity and reliance on tuning parameters necessitate careful evaluation against simpler methods on the specific dataset and task at hand.

The proposed model integrating pre processing, AFO with SVM based feature selection and support vector machine is a simple and efficient approach used to boost optimization capabilities without increasing computational complexity. On the other hand, it improves accuracy when compared to conventional procedures.

H. EXPERIMENTAL DESIGN

Google Colaboratory, an online integrated development environment, was used to test the algorithm. Google co-lab was set up on a Windows 10 PC equipped with a 2.4 GHz Intel Core i5 CPU and 4 GB of RAM. Numpy, Matplotlib, Scikit-learn, and Scipy are the libraries utilized.

I. DATASET

Researchers primarily utilized the Cleveland heart disease dataset from UCI for their heart disease prediction studies. Although it contains 303 records with 6 missing values and initially has 76 features, all published work focuses on a smaller subset of 13 key features, with the remaining feature solely indicating the presence or absence of heart disease. Table 4 illustrates the acquired dataset.

The Cleveland dataset is one of the most widely used benchmarks for CVD classification tasks. This allows for easy comparison of results with other studies and facilitates reproducibility. The data comes from actual patients, providing a realistic representation of the challenges and complexities encountered in clinical settings. The dataset is large enough to train and validate machine learning models but not so large as to be computationally expensive. It also includes a variety of relevant features, making it suitable for exploring different feature selection and model architectures. The dataset is publicly available, making it accessible to researchers and developers worldwide.

J. DATAPREPROCESSING

The data used for this analysis was directly extracted from Kaggle without any pre processing. It was organized into a comma-separated values file with readily available columns. Notably, all variables were complete (no missing values) and represented either continuous or categorical data. Interestingly, both categorical and numerical features in both heart disease and non-heart disease groups exhibited normal distributions. This suggests the data is potentially suitable for further analysis using statistical methods that rely on normality assumptions.

TABLE 4. Heart disease dataset.

Feature	Description
Age	Continuous variable describing the age of the patient
Sex	Categorical variable with two values- male, female
Cp	Categorical variable with 4 values Type1 is typical angina Type2 is atypical angina Type3 is non-anginal pain Type4 is asymptotic
Trestbps	Continuous variable Resting blood pressure in mmHg
Chol	Continuous variable showing serum cholesterol in mg/dl
Fbs	Categorical variable with 2 levels- true, false Fasting blood sugar>120mg/dl
Restecg	Fasting blood sugar >120mg/dl
Thalach	Categorical with 3 levels-N, L1, L2 Resting electrocardiographic results
Exang	Continuous variable describing maximum heart rate achieved
Oldpeak	Categorical with 2 levels-yes,no Exercise induced angina
Slope	Continuous variable describing ST depression induced by exercise relative to rest
Ca	Categorical with 3 levels-Up, flat, down The slope of the peak exercise ST segment
Thal	Categorical variable with 4 levels-0,1,2,3 Number of major blood vessels colored by fluoroscopy

K. OPTIMAL FEATURE SELECTION

This study aims to improve the accuracy of heart disease prediction by identifying key medical factors through feature selection. Choosing the most impactful features by AFO-SVM model from a larger pool not only reduces model complexity and data collection costs but also improves data quality, prediction performance, and computational efficiency. By leveraging the AFO-SVM technique, we were able to achieve impressive results in heart disease prediction using a minimal set of features. Compared to other feature selection methods that typically require more features, AFO-SVM identified only 6 key features out of 13, demonstrating its ability to achieve accurate predictions with greater efficiency.

Table 5 compares the number of features selected by different feature selection techniques for the Cleveland heart disease dataset. We can see that Relief, Info gain, Chi-squared, and Gain ratio all selected all 13 features, while Genetic Algorithm and its variants, PSO and AFO, selected fewer features (between 6 and 8). This suggests that the latter techniques may be more effective in identifying the most relevant features for this dataset, while the former techniques may be less efficient or may suffer from overfitting. Additionally, using SVM with some of the feature selection techniques led to further reduction in the number of features selected, indicating that these techniques may be complementary.

TABLE 5. Comparison of feature selection techniques.

Feature selection Technique	Dataset	Number of features selected
Relief	Cleveland	13(1,2,3,4,5,6,7,8,9,10,11,12,13)
Info gain	Cleveland	13(1,2,3,4,5,6,7,8,9,10,11,12,13)
Chi-squared	Cleveland	13(1,2,3,4,5,6,7,8,9,10,11,12,13)
Gain ratio	Cleveland	13(1,2,3,4,5,6,7,8,9,10,11,12,13)
Genetic Algorithm	Cleveland	7(3,7,8,9,10,12,13)
GA with SVM	Cleveland	7(3,7,8,9,10,12,13)
PSO	Cleveland	8(3,7,8,9,10,11,12,13)
PSO with SVM	Cleveland	7(3,7,8,9,10,12,13)
AFO	Cleveland	7(3,7,8,9,10,12,13)
AFO with SVM	Cleveland	6(3,7,8,9,10,13)

L. CLASSIFICATION WITH SVM

There are several compelling reasons why SVM is a popular choice for heart disease classification. Heart disease diagnosis often involves non-linear relationships between symptoms and features. SVM excels in recognizing these complex patterns, making it well-suited for this task. Real-world medical data can be noisy and contain outliers. SVM is robust to such imperfections, minimizing their impact on classification accuracy. Heart disease involves analyzing numerous features like blood pressure, ECG readings, and cholesterol levels. SVM can effectively handle high-dimensional data, making it ideal for this scenario. SVM generally selects only a small subset of the most relevant features for classification. This not only improves efficiency but also enhances interpretability, allowing physicians to understand the model's reasoning behind its predictions.

When compared with other algorithms like Random Forest (RF), K-Nearest Neighbor (KNN) and Multi-Layer Perceptron (MLP), SVM is excellent at handling non-linear data, robust to outliers, good for high-dimensional data. No single classifier dominates in all situations. Evaluating their effectiveness requires considering specific data, task, and desired characteristics of the model. SVM, with its non-linear capabilities and robustness, can be a strong contender, but its performance needs to be compared to other methods in the context of your specific application.

IV. PERFORMANCE EVALUATION

To assess the proposed algorithm's prediction accuracy, the study employs seven performance indicators based on the confusion matrix. The confusion matrix is used for this binary classification problem, highlighting how true and predicted values fall into four categories: True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN). Analyzing these categories within the confusion matrix allows for a comprehensive evaluation of the algorithm's effectiveness in predicting both positive and negative cases.

Based on the values in the confusion matrix, the six evaluation indicators can be calculated using these specific formulas. To evaluate the effectiveness of our AFO-SVM model for heart disease diagnosis, we employ Eq. (15), (16), (17), (18), (19), (20) and (21) to calculate critical metrics like RMSE, accuracy, precision, F-score, PME, specificity, and sensitivity, ensuring a comprehensive understanding of its strengths and limitations.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \tag{15}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{16}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\left(FP^2 + FN^2\right)} \tag{17}$$

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{18}$$

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{19}$$

$$\text{Sensitivity} = \frac{TP}{TP + FN} \tag{20}$$

$$\text{PME} = \frac{100}{TP + FP + TN + FN}\left(\frac{FP + FN}{TP + FN}\right) \tag{21}$$

The Receiver Operating Characteristic (ROC) curve is a powerful tool for assessing an algorithm's ability to generalize effectively. It plots the True Positive Rate (TPR) on the vertical axis and the False Positive Rate (FPR) on the horizontal axis. This visualization allows us to understand the trade-off between correctly identifying positive cases (TPR) and incorrectly identifying negative cases (FPR) at different threshold settings. By analyzing the curve's shape and area under the curve (AUC).

The AFO-SVM model addresses the critical need for medical information categorization by focusing on crucial features for better disease understanding. building highly accurate and reliable prediction models for informed treatment decisions. Providing interpretable results to empower physicians and enable personalized medicine. Offering an efficient and scalable solution for practical clinical use. By tackling these aspects, the AFO-SVM model has the potential to significantly improve the categorization of medical information and support effective and data-driven decision-making in cardiac treatment choices.

AFO-SVM has limitations that need to be addressed for optimal performance limitations. Depending on the problem size and the number of iterations, AFO can be computationally expensive compared to simpler algorithms. AFO-SVM relies on several parameters like number of flora and propagation distance, which require careful tuning for optimal performance. Improper tuning can lead to suboptimal results. In some cases, AFO-SVM might converge prematurely to suboptimal solutions, especially with smaller populations or inappropriate parameter settings. While AFO selects feature subsets, understanding the rationale behind their selection can be challenging.

AFO-SVM's Contributions to cardiovascular disease diagnosis and treatment. Accurate categorization of individuals at risk for CVD using AFO-SVM can lead to earlier diagnosis and intervention, improving patient outcomes. Patient-specific insights from feature selection can guide the

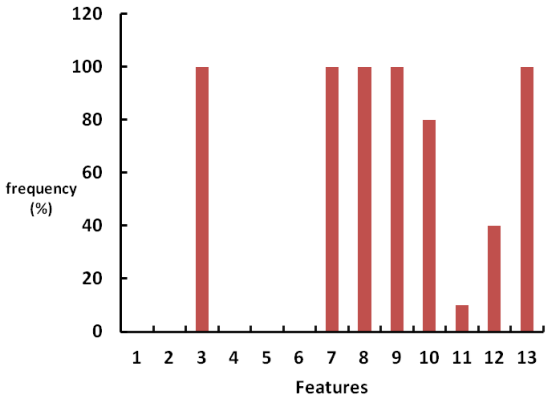
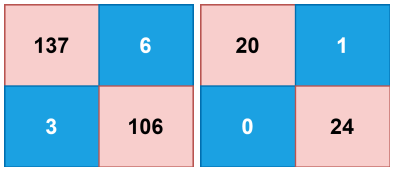


FIGURE 4. Frequency of features selected in K fold.



(a) Confusion matrix of training set (b) Confusion matrix of testing set

FIGURE 5. Confusion matrix.

development of targeted treatment plans, optimizing therapy effectiveness and reducing unnecessary interventions. Stratifying patients based on their AFO-SVM-derived risk profiles can facilitate the allocation of resources for preventive measures and early treatment, optimizing healthcare efficiency. Streamlined clinical decision-making: Interpretable AFO-SVM models can support physicians in making data-driven decisions, promoting evidence-based practice and reducing diagnostic uncertainty.

V. RESULTS AND DISCUSSION

This section delves into the findings and evaluation of the trained AFO-SVM approach for heart disease detection. K-fold cross-validation has been performed on the dataset with $k = 10$ for feature selection. The graph in Fig. 4 shows the frequency of features selected in each fold. The number of features and the features selected in each fold are represented in Table 6.

Utilizing diverse evaluation metrics, Tables 7 and 8 showcases the impressive performance and efficiency of the AFO-SVM-based technique. Table 7 shows the performance of the algorithm by presenting confusion matrix, which breaks down their prediction outcomes into percentages of True Positives, False Positives, True Negatives, and False Negatives.The confusion matrix for training set and testing set is given in Fig. 5.

Table 8 shows that the model performs well on both the training and test sets. On the training set, it correctly identifies 137 true negatives and 106 true positives, with only 3 false negatives and 6 false positives. On the test set, it performs even better with 20 true negatives and 24 true positives, with no false negatives and only 1 false positive. Overall, the model correctly classified 297 out of 302 individuals (98.31%). This indicates a promising performance for the model in identifying heart disease with high accuracy and low error rates. Table 8 summarizes the average performance of the AFO-SVM across key metrics like accuracy, sensitivity, and F-score, providing a comprehensive overview of its effectiveness.

Table 8 dives deeper into the performance of the heart disease prediction model, presenting a range of metrics beyond overall accuracy. RMSE (Root Mean Square Error) measures the average prediction error, with lower values indicating better accuracy. Both training and test sets show very low RMSE values (0.0357 and 0.0222), suggesting the model's predictions are very close to the actual values. Sensitivity captures the model's ability to identify true positives (people with the disease). Both sets show high sensitivity (99.2% and 97.25%), indicating the model rarely misses diseased cases. Specificity measures the model's ability to identify true negatives (healthy individuals). Again, both sets boast high specificity (95.82% and 95.24%), suggesting the model rarely misclassifies healthy people as diseased. Precision reflects the proportion of positive predictions that are actually correct (true positives out of all positive predictions). While slightly lower than sensitivity, both sets show high precision (96.0% and 94.64%), indicating the model rarely identifies healthy people as diseased. F-score combines precision and recall (sensitivity) into a single metric, balancing both aspects.

Both sets have high F-scores (0.9796 and 0.9593), demonstrating a strong balance between correctly identifying diseased and healthy individuals. PME (Partial Matching Error) measures the average difference between predicted and actual values. Unfortunately, the table only shows the test set value (3.57), but a lower value generally indicates better agreement. Accuracy represents the overall percentage of correct predictions. Both sets show high accuracy (97.78% and 96.43%), solidifying the model's effectiveness in predicting heart disease.Overall, this table paints a picture of a highly accurate and reliable model. The low error rates across various metrics suggest the model effectively identifies both healthy and diseased individuals with minimal mistakes. This reinforces the promising potential of this model for real-world applications in heart disease diagnosis and prediction.

While the SVM model achieved decent accuracy 79.35% for heart disease prediction using all available features, the AFO-SVM feature selection method significantly improved performance by identifying just six key features (sex, fasting blood sugar, maximum heart rate, exercise angina, actual, and number of main vessels). This targeted approach using the AFO-SVM classifier boosted accuracy to an impressive 96.63%, demonstrating its effectiveness in identifying heart disease with greater precision.

TABLE 6. Cross-validation results of feature selection by AFO-SVM.

K-Fold	Number of features selected	Features	RMSE (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F-Score (%)	PME (%)	Accuracy (%)
1	7	3,7,8,9,10,12,13	0.0362	97.28	97.20	94.29	97.03	3.53	96.15
2	6	3,7,8,9,10,13	0.0330	97.27	95.50	96.74	96.46	3.18	96.44
3	7	3,7,8,9,10,12,13	0.0369	97.98	95.80	94.88	96.17	3.03	95.52
4	6	3,7,8,9,11,13	0.0413	95.83	94.31	93.83	95.99	3.68	95.43
5	6	3,7,8,9,10,13	0.0325	96.02	95.19	95.71	94.81	3.89	97.44
6	6	3,7,8,9,12,13	0.0325	97.18	95.84	93.67	95.57	3.84	97.99
7	7	3,7,8,9,10,13	0.0416	96.73	94.58	95.10	95.83	2.95	96.56
8	6	3,7,8,9,10,12,13	0.0375	98.05	96.11	92.93	97.35	3.22	97.63
9	7	3,7,8,9,10,12,13	0.0314	96.83	95.13	93.56	96.63	3.54	96.99
10	6	3,7,8,9,10,13	0.0364	96.33	95.44	95.09	94.53	3.86	95.98

TABLE 7. The confusion matrix of AFO-SVM.

	TN	FN	TP	FP	Total
Train set	137	3	106	6	252
Test set	20	0	24	1	45
Overall	157	3	130	7	297

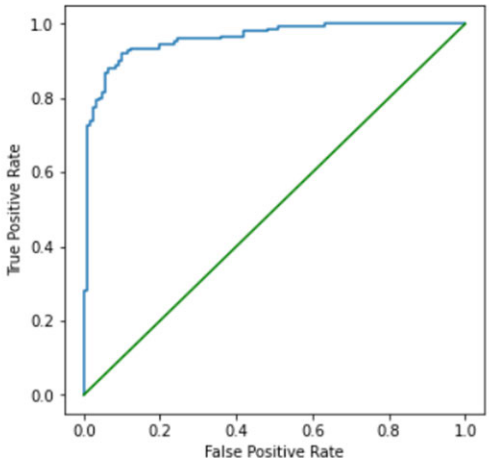


FIGURE 6. ROC curve for proposed classifier.

To demonstrate the superiority of our AFO-SVM approach, we compared it to established feature selection methods like Gain Ratio, Info Gain, Chi-square, GA, and PSO. Subsequently, we evaluated the SVM classifier's performance using Receiver Operating Characteristic (ROC) analysis as the gold standard. Fig. 6 showcases the outstanding ROC curve of the proposed model, further solidifying its effectiveness in heart disease prediction. The study presented a novel AFO-SVM feature selection method for heart disease prediction using the Cleveland dataset. To showcase the importance of its selected features, the SVM model trained on these features was compared to both established classification algorithms (RF, KNN, MLP) and other feature selection techniques (Info Gain, Gain Ratio, Chi-squared, Genetic Algorithm with/without SVM, PSO with/without SVM).

Table 9 showcases the performance of various machine learning models on the Cleveland heart disease dataset. Interestingly, both SVM and RF classifiers achieve the highest accuracy, surpassing 86%. [52] SVM model in 2020 edges out [49] 2021 Random Forest model with an accuracy of 87.9% compared to 86.9%. Meanwhile, KNN and MLP fall behind, showcasing accuracy scores of 80% and 81%, respectively. This comparison suggests that for the Cleveland dataset, SVM and RF algorithms might be particularly well-suited for accurate heart disease prediction.

Table 10 compares the performance of various feature selection methods and classification algorithms for heart disease prediction. Each row represents a different approach, with columns showing accuracy for four different classification models: RF, KNN, MLP, and SVM. The combination of AFO feature selection and the SVM classifier consistently outperforms all other methods, with an impressive overall accuracy of 96.63%. This suggests that AFO-SVM effectively identifies the most relevant features for accurate heart disease prediction using the SVM model. Across most feature selection methods, the SVM classifier generally achieves the highest accuracy. This highlights its potential as a powerful tool for heart disease prediction, especially when combined with effective feature selection techniques like AFO. While not reaching the peak performance of AFO-SVM+SVM, other combinations also achieve decent accuracies. Notably, GA+SVM and PSO+SVM show promising results, suggesting potential alternatives for feature selection and classification. Overall, this table provides valuable insights into the effectiveness of different approaches for heart disease prediction. While AFO-SVM emerges as the most promising combination, the table also demonstrates the potential of other techniques and the importance of finding the right match between feature selection and classification algorithms. The valid and important limitations to consider in the current work.

- Use additional datasets representing a wider range of data distributions and problem complexities. This will provide a more robust evaluation of the model's generalizability.
- Investigate and test other hybrid classifiers that might achieve better performance on the specific problem and dataset. Compare different approaches like Decision Trees, Random Forests, or XGBoost with SVM.
- Expand the comparison to include a wider range of relevant algorithms, especially those achieving strong

TABLE 8. The outcome of AFO-SVM based technique.

	RMSE	Sensitivity	Specificity	Precision	F-score	PME	Accuracy (%)
Train set	0.0357	99.2	95.82	96.0	0.9796	2.22	97.78
Test set	0.0222	97.25	95.24	94.64	0.9593	3.57	96.43
Over all	0.0337	97.74	95.73	94.89	0.9629	3.37	96.63

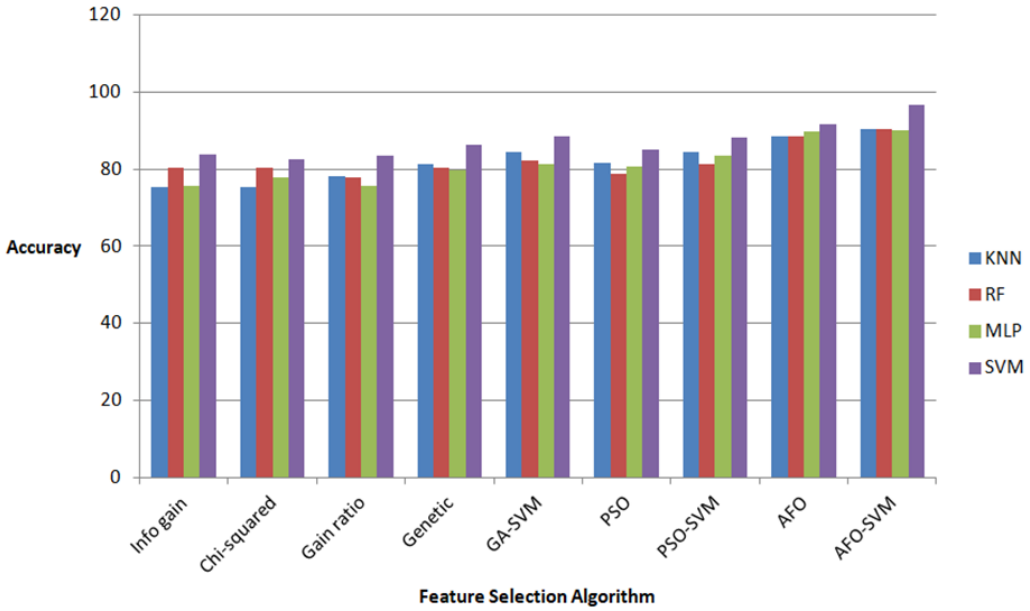


FIGURE 7. Accuracy comparison of various classifiers with AFO-SVM.

TABLE 9. Comparison of classifiers used.

Author	Dataset	Classifier	Accuracy(%)
Pal Madhumita [49]	Cleveland	RF	86.9
Khateeb, Nida [50]	Cleveland	KNN	80
Hasan TabreerT et al [51]	Cleveland	MLP	81
Sandhya Yamala [52]	Cleveland	SVM	87.9

TABLE 10. Accuracy comparison of various classifiers.

Methodology	RF(%)	KNN(%)	MLP(%)	SVM(%)
Info-gain	80.33	75.18	76.63	83.70
Chi-squared	80.24	75.18	77.75	82.50
Gain ratio	77.67	78.14	75.54	83.40
GA	80.26	81.4	79.81	86.22
GA+SVM	82.3	84.3	81.32	88.34
PSO	78.74	81.53	80.53	85.44
PSO+SVM	81.23	84.22	83.52	88.23
AFO	88.36	88.46	89.62	91.46
AFO+SVM	90.24	90.36	89.93	96.63

performance on similar problems. Research recent advancements in the field and include those algorithms.

- The dataset contains small number of samples, which is insufficient for building highly reliable and generalizable machine learning models, particularly in deep learning where larger datasets are typically required.Small sample sizes increase the risk of overfitting, leading to models that may perform well on the training data but fail to generalize to unseen data.
- The dataset lacks modern clinical and genomic markers that could improve predictive performance. There may be class imbalances between individuals with and without heart disease, as well as imbalances in key demographic variables such as gender and age.

VI. CONCLUSION AND FUTURE WORK

We analyzed heart disease data from the Cleveland dataset, preprocessed the data with z-score normalization, extracted relevant features using a novel AFO-SVM technique, and built a classification model using an SVM. Our model achieved exceptional performance, with an accuracy of 96.63%, specificity of 95.73%, sensitivity of 97.74%, precision of 94.89%, and F1-score of 96.29%. This suggests that AFO-SVM extracted key features for accurate heart disease detection using the SVM model.

The main contributions of this study can be summarized as follows.

- The study assessed the performance of popular machine learning algorithms like Random Forest, K-Nearest Neighbor, Multilayer Perceptron, and Support Vector Machine on a heart disease dataset to determine their effectiveness in predicting heart disease.
- The study also compared existing feature selection methods on the Cleveland dataset, and introduced a novel technique called AFO-SVM which achieved

superior performance with a remarkably small selection of just 6 features.

- The effectiveness of the proposed model was assessed using a range of metrics, including RMSE for error analysis, PME for performance evaluation, precision and specificity for classification accuracy, and sensitivity for true positive detection.

The proposed heart disease prediction model using AFO-SVM feature extraction demonstrates impressive robustness, efficiency, and individual feature analysis capabilities. Furthermore, its reliance on fewer features translates to lower computational cost compared to prior methods, and its straightforward classification scheme makes it highly promising for real-world clinical applications.

Despite the promising results, certain restrictions must be acknowledged. First, the study relied on a single dataset, which may not be applicable to different demographics or patient groups. Furthermore, the study only looked at a subset of demographic and clinical data, leaving out other potential risk factors for heart disease, such as lifestyle or genetic predispositions. Furthermore, the model's performance on a held-out test dataset was not examined, which would have offered information about how well the model generalizes to new, previously unseen data. Given these restrictions, further study is needed to overcome these difficulties and gain a better understanding of the potential of feature extraction for cardiovascular disease prediction. Therefore, future studies should address these limitations by:

- Including data from diverse populations and incorporating a broader range of clinical and demographic features to improve generalizability.
- Implementing rigorous testing methods like cross-validation or independent test datasets to assess the model's performance on new data.
- Exploring further feature extraction techniques: Investigating the effectiveness of feature extraction methods and exploring potential improvements.

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